5-MINUTE REVISION

Analysis of Covariance

From the chapter's own revision points, formulae and questions.

Selected from Chapter 22 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- ANCOVA combines treatment comparison with regression adjustment.
- A covariate should be relevant to the response and should not be caused by the treatment being compared.
- The common-slope model assumes the same regression slope across treatments; examine treatment-by-covariate interaction.
- Independent errors, appropriate linearity and common residual variance are required for the stated model; normal errors support the usual exact tests.
- Adjustment compares treatment means at a common covariate value.

Quick revision · 2

- Estimating one covariate slope uses one additional error degree of freedom.
- A useful covariate can improve precision, but improvement is not guaranteed.
- Adjustment does not turn an observational comparison into a randomized experiment.
- Use pair-specific standard errors when covariate means differ.

Important formulae · 1

For the CRD with one covariate and a common within-treatment slope:

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}, \qquad \hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

Here E_{xx}, E_{xy} and E_{yy} are the within-treatment corrected sums of squares and products defined in the chapter, with $E_{xx} > 0$.

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}, \qquad MSE_{adj} = \frac{E}{n - v - 1}$$

With n observed responses, v treatments and one fitted covariate slope, the CRD error degrees of freedom are $n - v - 1$. For the complete RBD with r blocks, use the within-block-and-treatment quantities and error degrees of freedom $(r-1)(v-1) - 1$.

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

$$F_{Treatment} = \frac{(E_1 - E)/(v-1)}{MSE_{adj}}$$

Important formulae · 2

Here E_1 is the residual sum of squares from the reduced model without treatment effects, retaining the covariate and, in RBD, the blocks. E is from the full model. The treatment comparison is therefore adjusted for the covariate.

Self-check from the chapter

1. ANCOVA combines analysis of variance with _____. **[Regression]**
2. A measured explanatory variable used for statistical adjustment is called a _____. **[Covariate]**
3. The common-slope ANCOVA model assumes equal regression _____ across treatments. **[Slopes]**
4. A covariate measured before treatment application is called a _____ covariate. **[Pre-treatment]**